Naveen Kumar Gadde *Renewable Energy Enthusiast*

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+49-1726012568

Munich, Germany

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ABOUT

Master's student in Hydrogen Technology with a completed Master's in Renewable Energy Systems. Strong theoretical foundation in hydrogen technology, with skills in data analysis and engineering design. Seeking roles in hydrogen technology, energy analysis, engineering design, or sustainability to apply knowledge in practical settings.

EDUCATION

Masters in Hydrogen Technology, ECTS:3.2 — Oct 2024 – present
Technische Hochschule Rosenheim — Rosenheim, Germany
Specialization: Sources and generation of Hydrogen Technologies, Carbon management strategies, Techno-economic analysis, and simulation, Quality management

Masters in Renewable Energy Systems, ECTS:2.1 — Oct 2019 – Aug 2023
University of Applied Sciences Nordhausen — Nordhausen, Germany
Specialization: Solar Energy, Wind Energy, Bioenergy Systems and LCA of Renewable energy systems

SKILLS

Research & Analysis: Qualitative & Quantitative Research, Data Collection, Techno-economic Analysis, LCA, Meta-analysis

Programming & Libraries: Python language, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Plotly, SQL, MATLAB, Excel

Data Visualization & Tools: Power BI, Microsoft PowerPoint, AWS (EC2, S3, Lambda), Git, GitHub

Engineering & Design: AutoCAD 2D/3D, SolidWorks, PV*SOL, Autodesk Inventor, CATIA V5, ANSYS

Soft Skills: Analytical Thinking, Time Management, Teamwork, Communication, Flexibility

Documentation & Writing: LaTeX, Overleaf, Technical Report Writing, Microsoft Word

PROFESSIONAL EXPERIENCE

Master Thesis – Meta-Analysis of Future Final Energy Consumption, Greenhouse Gas Emissions, and Hydrogen Energy Demand in India and the European Union — Jan 2023 – Aug 2023
ESA2 GmbH and Hochschule Nordhausen – University of Applied Sciences — Nordhausen, Germany

- Carried out a meta-analysis of diverse studies and scenarios (IEA, IRENA, WWF, Eurostat, etc.) covering historical trends to 2050 projections across key sectors including industry, transport, and buildings.
- Projected India to reach 30% renewable electricity by 2050 and the EU to exceed 50%, with bioenergy impact anticipated post-2030.
- Estimated hydrogen consumption to rise from less than 2% to over 21-22% of the final energy mix in both the EU and India by 2050.
- Anticipated that by 2050, 60% of hydrogen demand in India and the EU will be industrial, with green hydrogen potentially meeting 50%.
- Demonstrated hydrogen's potential in transport, projecting hydrogen and e-fuels to exceed 40% by 2050, with slower biofuel adoption post-2030.
- Evaluated normative frameworks, modeling approaches, energy policies, and decarbonization strategies aimed at achieving net-zero emissions by 2070.

Data Analysis/ Machine Learning Internship (Remote) — Oct 2022 – Dec 2022
Technocolabs Softwares — Noida, India

- Conducted credit risk analysis using Bandora's dataset, applying advanced feature engineering to improve predictive accuracy by 15%.
- Developed classification and regression models achieving 90% accuracy, and deployed them using Django and Streamlit with version control through GitHub.

Working Student - HyPat-Global H2 Potential Atlas Project — Jan 2022 – Aug 2022
ESA2 GmbH — Dresden, Germany

- Analysed more than 25 studies and their respective scenarios covering hydrogen potential, infrastructure, technology, economics, environmental impacts, social factors, policies, and innovation systems.

- Assessed hydrogen energy pathways, final energy mix, and GHG emission scenarios across 3 target years: 2030, 2040, and 2050.
- Consolidated scenario data into 1 structured Excel framework, enabling systematic comparison of hydrogen consumption and energy-system trends.
- Developed automated reports and calculation sheets across 3 time horizons, supporting deeper analysis of hydrogen energy consumption patterns.

TECHNICAL TRAINING

AutoCAD, CATIA V5 & ANSYS Oct 2016 – Mar 2017
Guntur, India

- **AutoCAD:** 2D drafting, 3D modelling, technical drawings, dimensioning and annotations.
- **CATIA V5:** 3D part modelling and assembly design.
- **ANSYS:** Basic engineering simulation and analysis.

PROJECTS

Techno-Economic Evaluation of Dimethyl Ether (DME) Production Feb 2026 – Jul 2026
Technische Hochschule Rosenheim

- Performed techno-economic analysis of a 50,000 t/y DME production plant using DOE and Towler cost-estimation methods.
- Estimated total capital investment at $4.315M based on detailed process and cost analysis.
- Modelled profitability and ±20% sensitivity, achieving a 21.0% gross profit margin and identifying variable and raw material costs as key drivers.

GitHub Portfolio Oct 2022 – present

- Built and maintained Python projects focused on Machine Learning and Data Analysis.
- Created 2D/3D designs using AutoCAD.

Construction of desulphurization rack, for a biogas plant Oct 2020 – Feb 2021
Hochschule Nordhausen – University of Applied Sciences

- Preparation of drawings using Auto-desk Inventor, research, and integration into Bio-gas plant.

Energy Data Analysis with Python May 2020 – Aug 2020
Hochschule Nordhausen – University of Applied Sciences

- Preprocessed raw data using Python (NumPy, Pandas), fixing formatting errors, removing duplicates, and filling missing values to reduce errors by 50%.
- Used Matplotlib, Seaborn, and Plotly for visualization, accelerating analysis by 35% and enabling faster pattern and trend identification.

Energy Autonomous Pipette Oct 2019 – Feb 2020
Hochschule Nordhausen – University of Applied Sciences

- Explored powering an electronic pipette using integrated solar cells optimized for low indoor irradiance conditions.
- Measured energy consumption, indoor irradiance, and solar cell IV-curves, reaching an efficiency of almost 30%.

CERTIFICATES

Master Carbon Accounting & ESG Reporting [2025 CSRD Update]: CSRD, GHG Scopes 1–3, ESG, TCFD, GRI, ISO 14064

Python for Data Science and Machine Learning Bootcamp: Python, Pandas, NumPy, SQL, Data Visualization

Power BI - Business Intelligence for Beginners to Advance: Power BI, DAX, Power Query, ETL

PVSOL Premium Bootcamp : From A to Z: Solar PV Design, Optimization, Energy Yield Analysis

The Complete SQL Server: SQL, CTEs, Window Functions, Database Design

Project Management Foundations: Agile, Waterfall, Risk & Stakeholder Management

AWS Certified Solutions Architect – Associate: AWS Cloud, EC2, S3, IAM, VPC

LAB & FIELD EXPERIENCE

- Solar PV: Solar panel testing, IV-curve measurement, indoor irradiance testing

LANGUAGES

- English (Business Level)
- German(B1 Level)

Employment certificate / reference – ESA² GmbH

Mr. Naveen Kumar Gadde was employed as a working student at ESA² GmbH in the period from January 01, 2022 to August 31, 2022.

The ESA² GmbH understands itself as an innovative, independent, and competent consulting and research company for energy topics, which scientifically supports decision makers from politics and economy on the way of the energy transition.

During his work as a working student, Mr. Gadde was entrusted with the following tasks:

HYPAT project [joint project, Federal Ministry of Education and Research]:
Global H2 potential atlas - sustainable locations in the world for the green hydrogen economy of tomorrow.

- Recherche on current scenario studies on global energy demand development in general and on hydrogen demand development in particular (focus on the European Union and India)
- Evaluation of scenario data in the context of a meta-study (master thesis)

Mr. Gadde has good specialist knowledge, which he used confidently and successfully to complete his tasks. He was always an extremely motivated employee who approached the tasks set for him with great verve and found meaningful and practicable solutions.

Even in situations with a high workload, Mr. Gadde proved to be a very resilient employee. In doing so, he proceeded in a considered, calm, and goal-oriented manner at all times. In this way, he always delivered high-quality results.

In summary, Mr. Gadde always completed the tasks assigned to him to our satisfaction. His conduct towards superiors and external parties was always excellent.

We thank Mr. Gadde for the very good cooperation. We wish him all the best and continued success in his future career and life.

Dresden, August 08, 2023

Dr. Robert Kunze
Managing Director ESA² GmbH

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bearbeitet:
Nordhausen, 14. August 2023

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Naveen Kumar Gadde

Provisional certificate for the Master's examination

Dear Naveen Kumar Gadde,

I am enclosing the certificate of the successfully completed Master's examination with an overview of the examinations and academic achievements within the scope of your studies, so that you have meaningful documents for upcoming applications.

If you would like to have additional achievements listed on the certificate, please inform us immediately so that we can prepare the certificate.
Thank you very much.

Yours sincerely

HOCHSCHULE NORDHAUSEN
University of Applied Sciences

Bescheinigung über die Masterprüfung

Naveen Kumar Gadde
geboren am 2. Februar 1994 in Kanigiri

hat die Masterprüfung im Studiengang

Renewable Energy Systems (M.Eng.)

mit der Gesamtnote gut (2,1) bestanden

Modulprüfungen Pflichtbereich	Note	ECTS-Credits
Bioenergy Systems I (854)	befriedigend (2,7)	5
Photovoltaic Systems (852)	gut (1,7)	5
Wind Power Plants (851)	gut (2,3)	5
1st Scientific Project (857)	gut (1,7)	5
German as a foreign Language (910) A2 GER	befriedigend (3,0)	5
Bioenergy Systems II (855) - Biogas and Liquid Biofuels	gut (2,3)	5
Solar Thermal Lab (859)	gut (2,0)	5
Lifecycle Analysis of Renewable Energy Systems (856)	befriedigend (2,7)	5
2nd Scientific Project (858)	gut (1,7)	10

Modulprüfungen (Wahlpflichtbereich)	**Note**	**ECTS-Credits**
Elective Course - Fuel Cell Technologies (863)	gut (1,7)	5
Elective Course - Ocean Energy and Hydropower (861)	sehr gut (1,3)	5

Die schriftliche Masterthesis und das Kolloquium wurden über das folgende Thema abgelegt:

Meta-analysis an future hydrogen demand in India and the European Union

	Note	**ECTS-Credits**
Master Thesis	gut (2,3)	26
Masterkolloquium mit Rigorosum	gut (2,3)	4

Gesamtumfang vorgenannter Pflichtleistungen 90 ECTS

Nordhausen, 7. August 2023

Siegel

i.V. Beate Dietrich
Sachgebietsleiterin
Studien-Service-Zentrum

Technocolabs Softwares Inc.

Letter of Recommendation

Date : 30-12-2022

To Whom It May Concern,

With immense pleasure, I am writing this to recommend ***Mr. Naveen Kumar Gadde.*** I have known him in my capacity from ***01st October 2022 to 15th December 2022*** and he successfully completed this remote internship as a Machine Learning Engineer Intern at Technocolabs, Indore. During the internship, he was found to be proactive, punctual, and inquisitive and demonstrated a good work ethic and interpersonal skills. ***Mr. Naveen Kumar Gadde*** carried out all the tasks that were assigned by the company and he made sure to put in extra hours of effort and completed the work within the mentioned deadlines.

Our team grouped him up with a few other interns to complete the company's major project and we found out that ***Mr. Naveen Kumar Gadde*** exhibited profound team management and communication skills that helped him throughout the completion of the project. Besides this, he gained experience in numerous concepts in Machine Learning that enhanced his existing skillset. I am impressed by his professionalism and dedication. I definitely believe that he would be an excellent fit and will bring great laurels to any university or company that he applies to, in the near future.

We wish ***Mr. Naveen Kumar Gadde*** all the very best and success in his future endeavors.

If you've any questions regarding the recommendation, please feel free to contact me or mail me regarding the same.

Yours Sincerely,



Name of the recommender: **Yasin Shah**
Designation: **Founder and Chief Executive Officer, Technocolabs**
Email id: contact@technocolabs.com / shahyaseen71@gmail.com
Contact Number: +91 8319291391
City: Indore, Madhya Pradesh, India.

Technische Hochschule Rosenheim

Technische Hochschule Rosenheim – Technical University of Applied Sciences
Hochschulstraße 1, 83024 Rosenheim

Herrn

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84489 Burghausen

Präsident
Prof. Heinrich Köster

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Verwendungszweck 1542

Notenbestätigung / Transcript Of Records

Vorname, Nachname First name, surname	**Gadde Naveen Kumar**
Geburtsdatum und -ort Date and place of birth	**02. Februar 1994** **Kanigiri**
Studiengang Study course	**Masterstudiengang Hydrogen Technology** **Master's Degree Programme Hydrogen Technology**

hat folgende Module abgelegt:
has completed the following courses

Module/Moduls		Note/ Grade	Semester	ECTS
Vorleistungen **Summary of ECTS**		**BE**	**WiSe 2024/25**	**210**
HYT-M 20242 VS	Vorstudium First Academic Degree	BE	WiSe 2024/25	210 ****
X-0 0 VS.M	Vorstudium First Academic Degree	BE	WiSe 2024/25	210
Vorstudium **First Academic Degree**				**210**
Gesamtkonto **Summary of ECTS**		**3,2**	**SoSe 2026**	**25**
Pflichtmodule **Compulsory Modules**				**5**
HYT-M 20242 01000.M	Wasserstoff: Grundlagen und Sicherheit Fundamentals of Hydrogen and Safety	4,0	WiSe 2025/26	5

Module/Moduls		Note/ Grade	Semester	ECTS
Fachwissenschaftliche und Applikations- & kompetenzorientierte Wahlpflichtmodule **Specialization and Application & Competence Oriented Elective Courses**				**20**
HYT-M 1 10800.M	Techno-Economic Analysis and Simulation Techno-Economic Analysis and Simulation	1,7	SoSe 2026	5
HYT-M 1 10600.M	Hydrogen Storage, Transportation and Distribution Systems Hydrogen Storage, Transportation and Distribution Systems	3,0	WiSe 2025/26	5
HYT-M 1 10110.M	Introduction to the Economics of Hydrogen Markets Introduction to the Economics of Hydrogen Markets	3,7	SoSe 2026	5
HYT-M 1 10160.M	Carbon Management Carbon Management	4,0	SoSe 2025	5

Legende:

AWPM = Allgemeinwissenschaftliches Wahlpflichtmodul/*General required Elective Courses*, BA = Bachelorarbeit/*Bachelor's thesis*, ECTS = European Credit Transfer System, elP = elektronische Prüfung/*electrical examination*, Ex= Exkursion, FWPM = Fachbezogenes Wahlpflichtmodul/*Specialist required Elective Courses*, Kol = Kolloquium/*colloquium*, LNmE = Teilnahmenachweis mit Erfolg/*proof of participation with success*, MA = Masterarbeit/*Master's thesis*, mdlP = mündliche Prüfung/*oral examination*, mE = mit Erfolg abgelegt/*pass*, P = Prüfungen/*examination*, PA = Projektarbeit/*project work*, PB = Praxisbericht/*practice report*, Pr= Praktikum/*work experience*, prP = praktische Prüfung/*practical examination*, PStA = Prüfungsstudienarbeit/*coursework*, S = Seminar/*seminar*, schrP = schriftliche Prüfung/*written examination*, SU = Seminaristischer Unterricht/*seminar-based lectures*, SV = Seminarvortrag/*seminar presentation*, SWS = Semesterwochenstunden/*hours per week per semester*, TN = Teilnahmenachweis/*attendance*, Ü = Übung/*practical exercise*, V = Vorlesung/*lecture*, ZV = Zulassungsvoraussetzung/*admission requirements*

NZ = Nachträgliche Zulassung/*subsequent admission to examination*, ZU = Angemeldeter Leistungsnachweis/*registered performance record*, F1 = Nicht fristgerecht abgelegte Prüfung (Erstversuch)/*examination not taken on time (first attempt)*, F2 = Nicht fristgerecht abgelegte Prüfung (Zweitversuch)/*examination not taken on time (second attempt)*, F3 = Nicht fristgerecht abgelegte Prüfung (Drittversuch)/*examination not taken on time (third attempt)*, NA = Nichtabgabe/nicht abgelegt/*not handed in/not taken*, F1 GOP = Grundlagen- und Orientierungsprüfungen nicht fristgerecht abgelegt/*basic and orientation examination not taken on time (first attempt)* , Ä = Notenänderung/*grade change,* Gew. = Gewicht/*weighting*

* anerkannt/*accredited*
** Fristen zur Ablegung der Wiederholungsprüfung werden durch die Beurlaubung oder Exmatrikulation nicht unterbrochen.
*** Die Noten des aktuellen Semesters werden erst mit dem Tag der Notenfeststellung durch die Prüfungskommission verbindlich und veröffentlicht. Diese findet am 14.2. im Wintersemester und am 31.7. im Sommersemester statt. Sofern dieser Tag ein Wochenende oder Feiertag ist, fand die Sitzung am vorherigen Werktag statt.
**** Die erzielten ECTS gehen nicht in die Gesamtnote ein./*The ECTS achieved do not count towards the overall grade.*
EN = endgültig nicht bestandenen / *irrevocably failedirrevocably failed*

International Grade System (ECTS = European Credit Transfer System)

ECTS – Grade	German Grade	ECTS – Definition	Translation
A	1,0 – 1,5	Excellent	hervorragend
B	1,6 – 2,0	Very Good	sehr gut
C	2,1 – 3,0	Good	gut
D	3,1 – 3,5	Satisfactory	befriedigend
E	3,6 – 4,0	Sufficient	ausreichend
FX/F	4,1 – 5,0	Fail	nicht bestanden

Druckdatum: 02/08/2026

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Student Administration

Technische Hochschule **Rosenheim**

Präsident
Prof. Heinrich Köster

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Mr.
Gadde Naveen Kumar
In den Grüben 190
84489 Burghausen

Enrolment certificate

also: certification in accordance with Section 9 BAföG

Mr.	Gadde Naveen Kumar
Registration number	1103784
Date of birth	02/02/1994
Place of birth	Kanigiri, India
Is enrolled at	Technical University of Applied Sciences Rosenheim
in	winter term 2026/27
	and currently not on leave of absence.
First-time enrollment at a university	winter term 2019/20
Semester duration	10/01/2026 - 03/14/2027
Course of study	Master's Degree Programme Hydrogen Technology
University semester (1)	13
Curricular semester (2)	3
Subject semester (3)	5
Type of study	full-time
Standard period of study	3 semester
Examination regulation	ER WiTe 2024/25
Student status	student
Attained ECTS / as of	25 / 08/21/2026

(1) University semesters correspond to the number of semesters completed in Germany. | (2) Semesters of taught content document the academic progress. This information is subject to the final classification to a semester in terms of the respectively valid study and examination regulations. | (3) The semester shows how long the current degree programme is taking.

The standard period of study indicated on this certificate is being extended respectively by one semester on account of Art. 99 (2) of the Bavarian Higher Education Act (BayHSchG), and a new individual standard period of study is therefore specified.

Degree-specific comments can be found in the respective Study and Examination Regulations.

Stand: 08/21/2026

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Diploma Certificate

Regd. No. 575/93

The A CARE EXAMINATION BOARD Certifies that GADDE NAVEN KUMAR, S/o SRINIVASA RAO *Reg No.* GNF060105

has qualified in AUTOCAD-CATIAYA V5-ANSYS,

Course as per the Board requirement from A CARE GUNTUR

Centre during the period OCTOBER - 2016 *to* MARCH - 2017

The Candidate secured FIRST *class with* 'A' *grade in*

the qualifying Examination held on 20-03-2017

Marks	Grade	Class
above 80%	A+	Distinction
60% to 80%	A	1
50% to 59%	B	2
35% to 49%	C	3

Course in-charge

ACARE

President

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Secretary

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